Understanding public health compliance in emergencies

An investigation of the heterogeneity and pathways of factors that shape our behaviour during public health crises in LMICs

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Introduction

Research Question

How do `dynamic interactions` among diverse behavioural determinants shape public health `compliance` during infectious disease outbreaks in LMICs?

Relevance: Global & Policy Relevance

Pandemic Impact in LMICs

- COVID-19 exposed complex challenges in public health compliance, `especially in low- and middle-income countries`.
- LMICs face unique socio-economic and institutional hurdles not fully captured by HIC-focused research.
- A better understanding of behavioural drivers is critical to design effective, context-specific public health interventions.

Relevance: Enabling better policy making

Predicting policy sucess instead of only ex-post evaluations

- Policymakers require robust, causal evidence to balance health protection with economic stability.
- Identifying the causal pathways in compliance
  - helps avoid unintended policy consequences
  - enables faster and more promising decision making during crises.

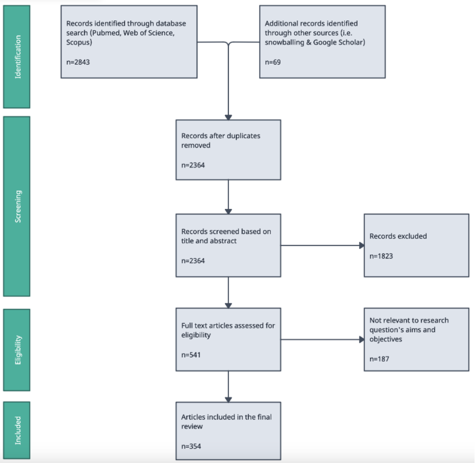
  Is there another pandemic coming? Yes. When? Which pathogen? How severe will it be? No one can say for sure.

Yonatan Grad, `Harvard T.H. Chan School of Public Health`

What's known? Literature Review

Literature search strategy

- Searched WOS, Scopus, Pubmed, G-scholar
- Added sources through snowballing
- 354 studies reviewed



Literature search strategy II: Search string (Skip)

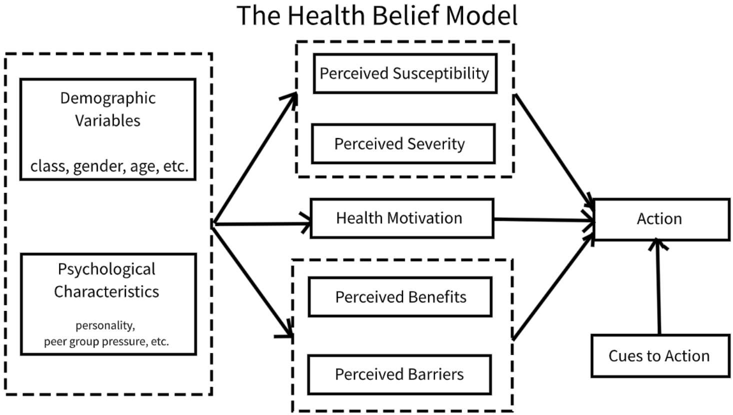
```
TITLE-ABS-KEY(
  (pandemic* OR epidemic* OR outbreak* OR "public health cr
```

Theoretical Frameworks that (don't) connect these factors

Frameworks: Theory of plnned behaviour

Figure 1: REFERENCE/SOURCE

Frameworks: Health Belief Model

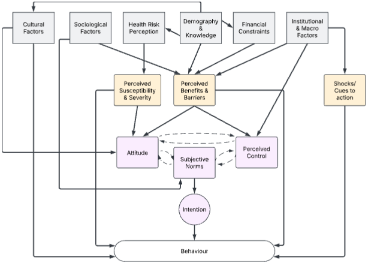


Chapter I: Comparative structural modelling to inform a theory of pandemic behaviour

RQ: What are the core mechanisms and interactions through which economic, socio-cultural, institutional, and psychological determinants shape individual compliance with public health measures during pandemics in LMICs?

Background & Gaps

- Some studies use theoretical frameworks (HBM `OR` TPB) to analyse survey data
- Only REFERENCE in LMIC (Looking at students in Turkey)
- No paper integrating/ extending these in C19 in LMIC
- No study comparing



Chapter II: Causally testing and qunatifying the health-income trade-off assumption

RQ: What is the causal impact of economic constraints relative to health risk perceptions and vice versa on compliance with public health measures in Malawi?

Background & Gaps

Widely assumed trade-off that `individuals` face

- Rationale behind cash transfer programmes
- Supported by economic theory (Grossman (19X); Prospect Theory)
- Empirical eveidence shows
  - Effects of income
  - Effects of risk perceptions
- Income and risk perception never modelled together to confirm `trade-off` in causal way

Hypotheses

H1(2): Economic vulnerability (health risk perceptions) predict

Chapter: Peer effects vs self selection

RQ: What is the causal effect of membership in non self-selected peer groups on adherence to public health measures during pandemics?

Background & Gaps

Methods

Data

Why: Contributions

Emprical contribution

Theoretical Contribution

Policy relevance

Thank you!

Questions & Feedback

Annex I: Chapter I

Annex: Structural Equation Modelling II (Skip)

Confirmatory Factor Analysis (CFA)

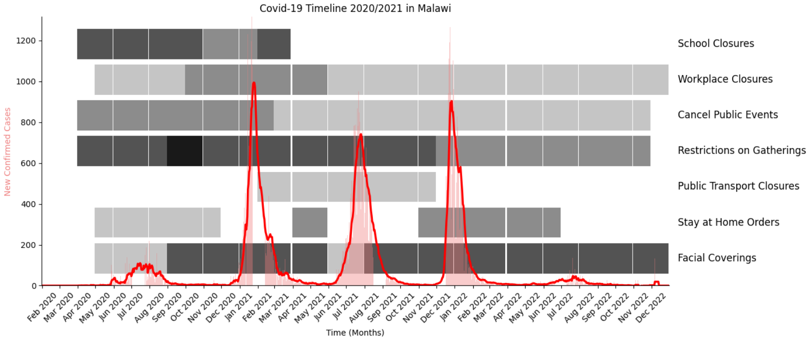
- To validate latent constructs
- To operationalise constructs from survey items

Comparative Model Evaluation

- Predictive power assessment (variance explained in preventative behavior)
- Multi-group SEM for cross-country/SES/gender comparisons
- Akaike Information Criterion (AIC) & Bayesian Information Criterion (BIC) for model comparison

Annex II Chapter 2

Annex: Chapter II BG: Malawi during waves 1 & 2 of C19



Annex: Chapter II How: Outcomes

Outcome	Indicators	Investigation at	Hypothesis tests
Social distancing	Gathering attendance; non-essential travels; Reduced Shopping	Multiple rounds	H1, H2, H3

Annex III Chapter 3